1 Introduction

The rapid expansion of environmental, social, and governance (ESG) reporting has made corporate environmental disclosure an important information channel for investors, regulators, and other stakeholders. Environmental communication can enhance legitimacy and attract capital, but these benefits also create incentives for firms to present a favorable environmental image without achieving commensurate environmental performance (Delmas and Burbano, 2011). Greenwashing therefore arises when environmental communication creates a strategic divergence from the substantive environmental performance that could support that communication (Lyon and Maxwell, 2011; Zhou et al., 2024). This divergence reduces the credibility and decision usefulness of environmental information because stakeholders may interpret favorable narratives as evidence of environmental conduct that the firm's actual performance does not support (Christensen et al., 2021; Zhou et al., 2024). An important but unresolved question is whether the ordinary financial-statement audit can constrain this communication–performance divergence even when no separate sustainability assurance is provided.

Environmental narratives in annual reports generally fall outside the direct scope of the financial-statement audit opinion. However, they are not necessarily isolated from the audit process. ISA 720 and its Chinese counterpart, CAS 1521, require auditors to read other information and consider whether it is materially inconsistent with the audited financial statements or with knowledge obtained during the audit. These requirements do not make the financial-statement auditor an assurer of environmental information. They nevertheless create a limited institutional channel through which particular environmental statements may be compared with information obtained during the ordinary audit. The possibility of such comparison depends on the content of the environmental statement. General aspirations, policy slogans, and broad commitments usually provide few concrete points of comparison. By contrast, statements describing environmental projects, technology adoption, capital investment, pollution remediation, equipment installation, regulatory rectification, or measurable environmental outcomes may contain factual anchors that have financial, operational, or compliance-related implications (Clarkson et al., 2004, 2008; Barth and McNichols, 1994). Such statements may therefore intersect with information about assets, expenditures, cash flows, production processes, environmental obligations, penalties, or other matters encountered during the audit. Accordingly, concrete action claims are more likely than symbolic claims to provide information that can be compared with the auditor's existing knowledge of the client and its industry.

Audit firm industry specialization may make this comparison more informative. Industry-specialist auditors accumulate knowledge about clients' business models, production processes, technologies, regulatory environments, and typical financial relationships (Solomon et al., 1999; Francis, 2011; Reichelt and Wang, 2010). This knowledge can help them recognize when several individually ambiguous signals form an unusual pattern, assess the risk associated with that pattern, and select more targeted follow-up procedures (Owhoso et al., 2002; Low, 2004; Hammersley, 2006). Applied to environmental narratives, industry expertise may improve the probability that an auditor recognizes a concrete environmental claim that is difficult to reconcile with the client's operations, financial information, or information obtained during the audit. If a potential material inconsistency becomes salient, CAS 1521 provides a basis for discussing the matter with management, considering whether further work is necessary, and communicating an uncorrected material misstatement to those charged with governance. They may increase the expected cost to management of maintaining action-oriented environmental claims that lack adequate support. This reasoning leads to our research question: is audit firm industry specialization associated with less corporate greenwashing?

We examine this question using 8,129 firm-year observations for 2,669 Chinese A-share listed firms from 2021 through 2024. We measure greenwashing as the within-industry-year divergence between the substantive share of environmental communication and an external environmental-performance score. The substantive share is identified using LLM-enabled semantic classification of annual-report text, following the DeepGreen procedure (Xu et al., 2026). We measure audit firm industry specialization using the audit firm's share of client operating revenue within an industry-year. The analysis controls for financial, governance, and audit characteristics and includes year and industry fixed effects. The main results show that audit firm industry specialization is negatively associated with corporate greenwashing. The relation remains negative as controls and fixed effects are introduced and is directionally consistent across alternative specialization measures and standard-error treatments.

This study contributes to the literature in three ways. First, it extends research on audit firm industry specialization and client disclosure quality by examining whether industry expertise is associated with the gap between environmental communication and environmental performance (Balsam et al., 2003; Dunn and Mayhew, 2004; Reichelt and Wang, 2010). Second, the study adds to greenwashing research by identifying a potential governance spillover from the ordinary financial-statement audit, rather than focusing only on firms' legitimacy incentives and external environmental scrutiny (Delmas and Burbano, 2011; Lyon and Maxwell, 2011; Marquis et al., 2016). Third, the study uses semantic classification to distinguish substantive from symbolic environmental communication in a large-sample auditing setting (Baier et al., 2020; Bilinski, 2024; Xu et al., 2026).

The remainder of the paper develops the theoretical arguments, explains the measurement and research design, presents the empirical results, and concludes.

2 Literature Review and Hypothesis Development

2.1 Greenwashing as an Action–Performance Divergence

Environmental disclosure can reduce information asymmetry by helping investors and other stakeholders assess firms' environmental responsibilities, exposures, and performance (Healy and Palepu, 2001; Verrecchia, 2001). Its usefulness depends on whether environmental communication provides a credible representation of underlying corporate conduct. When firms use favorable environmental communication strategically, disclosure can obscure rather than clarify environmental performance. Greenwashing is therefore commonly understood as a mismatch between positive environmental communication and the substantive environmental performance that could support that communication (Delmas and Burbano, 2011; Lyon and Maxwell, 2011).

Stakeholder expectations and legitimacy pressures increase the benefits of appearing environmentally responsible, whereas substantive environmental improvements may require costly changes in technology, production, and organizational practices (Delmas and Burbano, 2011). Firms may consequently rely on selective, exaggerated, or symbolic communication when outsiders cannot readily observe the underlying performance. Prior research shows that institutional norms, external scrutiny, and the expected cost of having environmental claims challenged can affect firms' disclosure choices (Kim and Lyon, 2015; Marquis et al., 2016; Lyon and Maxwell, 2011). Greenwashing is thus more attractive when the benefits of a favorable environmental image exceed the expected cost of an unsupported or incomplete narrative.

Our measure captures a communication–performance divergence that occurs when action-oriented environmental communication is relatively high but externally assessed environmental performance

is relatively low. This distinction is important because concrete environmental claims may be more informative and more readily comparable than general aspirations, even though they can still be exaggerated, selectively presented, or insufficiently supported.

2.2 Environmental Action Claims as Points of Comparison

Environmental disclosure research distinguishes between relatively concrete or "hard" information and broader "soft" statements about policies, values, and aspirations (Clarkson et al., 2008; Hummel and Schlick, 2016). Hard environmental disclosure may describe environmental investments, management systems, performance indicators, quantitative outcomes, environmental expenditures, environmental incidents, or progress toward specified targets (Clarkson et al., 2008). Soft disclosure, by contrast, may consist of general policies, principles, commitments, and favorable descriptions that provide fewer details against which the underlying conduct can be assessed (Clarkson et al., 2008; Hummel and Schlick, 2016). This distinction provides the first link between environmental communication and the ordinary financial-statement audit.

Concrete claims about environmental projects or operational actions may have observable financial or operational implications. For example, an environmental equipment project may relate to capital expenditures, construction in progress, procurement, or payments. Pollution remediation may relate to expenditures, environmental liabilities, provisions, or contingencies (Barth and McNichols, 1994; Barth et al., 1997; Li et al., 1997). Environmental technology upgrades may also be connected to equipment changes, production processes, operating costs, or industry-specific technical conditions (Clarkson et al., 2004). These relationships mean that action-oriented environmental statements are more likely than symbolic statements to create points of comparison with financial statements and with knowledge obtained during the audit.

CAS 1521 establishes the relevant procedural boundary. When reading other information, the auditor considers whether it is materially inconsistent with the financial statements or with knowledge obtained during the audit and remains alert to indications that the other information may be materially misstated. If an apparent material inconsistency arises, the auditor discusses the matter with management and considers whether additional procedures or communication are necessary. Accordingly, CAS 1521 does not require auditors to verify all environmental claims or to provide sustainability assurance. It does, however, provide a limited channel through which a concrete environmental claim may become relevant to the auditor when that claim intersects with available financial, operational, or compliance-related information.

2.3 Industry Specialization and Recognition of Inconsistencies

Industry specialization strengthens the information base relevant to this potential comparison. Through repeated engagements, specialist audit firms develop knowledge about industry business models, production processes, regulatory requirements, technologies, cost structures, and reporting risks (Francis, 2011; Reichelt and Wang, 2010). This knowledge helps auditors form more precise expectations about the relationships among a client's operations, financial information, and industry conditions. Experimental evidence indicates that industry specialists possess richer industry knowledge and are better able to identify errors or unusual patterns in industry-related information (Solomon et al., 1999; Owhoso et al., 2002). Industry specialization also affects audit risk assessments and planning decisions, including judgments about the nature and quality of subsequent audit procedures (Low, 2004). Hammersley (2006) further shows that industry specialists can identify complex error patterns formed by several individually ambiguous cues and can propose

more discriminating follow-up procedures (Hammersley, 2006).

These studies support a inference about comparative recognition ability. When an environmental action claim, a financial reporting item, and an operational or regulatory fact jointly form an unusual pattern, industry specialists may be better positioned to recognize that the pattern warrants attention. For example, an assertion that a major environmental upgrade has been completed may be more informative to an industry specialist who understands the technology, production scale, expected investment, and operational consequences associated with that upgrade. Similarly, a claim of comprehensive environmental rectification may be easier to evaluate when the auditor understands the client's exposure to industry regulation and the operational implications of the alleged rectification. The relevant mechanism is that concrete claims increase the opportunity for comparison, while industry specialization improves the auditor's ability to interpret whether the available information forms a potentially inconsistent pattern.

2.4 From Recognition to Informed Questioning

Recognition of a potential inconsistency may affect the information environment surrounding management's disclosure choices. Under CAS 1521, an auditor who identifies an apparent material inconsistency discusses the matter with management, considers the explanations and supporting information provided, and determines whether further action or communication is necessary. Industry knowledge can make these questions more specific because the auditor can relate management's explanation to the client's production activities, financial reporting, technology, and regulatory setting. More informed questioning implies that management may face greater difficulty sustaining a concrete claim that is difficult to reconcile with information already available to the auditor.

This reasoning is consistent with evidence that auditing can influence disclosure choices beyond the audited financial statements and that auditor industry specialization is associated with client disclosure quality (Schroeder, 2016; Dunn and Mayhew, 2004). A firm may still disclose environmental actions for legitimacy or capital-market reasons, but an industry specialist may increase the expected cost of presenting unsupported action claims as evidence of strong environmental performance. The resulting prediction is an association between specialization and less greenwashing.

2.5 Scope Conditions and Hypothesis

Several considerations limit this predicted association. The auditor's responsibility for other information is narrower than the responsibility arising in a separate sustainability assurance engagement. Financial-statement audit expertise may also be insufficient to evaluate the scientific or technical substance of some environmental claims. Some environmental actions may be implemented through arrangements that have no clear or material accounting manifestation. Firms also retain strong legitimacy and capital-market incentives to maintain favorable environmental narratives (Delmas and Burbano, 2011; Healy and Palepu, 2001; Verrecchia, 2001). Dedicated sustainability assurance remains a distinct governance mechanism with different procedures, expertise, and responsibilities (Simnett et al., 2009; Pflugrath et al., 2011; Cohen and Simnett, 2015). These limitations imply that industry specialization should not be expected to eliminate greenwashing or to affect every environmental statement. They do not, however, rule out a detectable average association when firms' concrete environmental claims provide points of comparison and specialist auditors possess a comparative advantage in recognizing inconsistent patterns. We therefore state the following hypothesis:

H1: Audit firm industry specialization is negatively associated with corporate greenwashing.

3 Research Design

3.1 Sample Selection and Data Sources

Our initial sample consists of Chinese A-share listed firms from 2021 through 2024. We exclude firms designated as *ST, ST, or PT during the sample period, as these firms may exhibit abnormal financial conditions and operating characteristics. Annual reports are obtained from the CNINFO website, and financial, corporate-governance, audit, and full-market audit-client data are primarily obtained from the China Stock Market & Accounting Research (CSMAR) database. We use Huazheng environmental scores as the external measure of firms' environmental performance. The annual-report text and environmental scores are used to construct the greenwashing measure, while the full-market audit-client data are used to calculate audit firm industry specialization. Financial, governance, and audit data provide the control variables. Following the industry-classification convention commonly applied to Chinese listed firms (Gul et al., 2010), we classify manufacturing firms at the CSRC two-digit industry level and firms in other sectors at the CSRC one-digit industry level.

The sample selection reflects both textual validity and data availability. We first retain firm-years for which the relevant annual-report text can be extracted completely and the green-disclosure identification procedure can be completed. Firm-years with extraction or classification failures are treated as incomplete rather than as having zero green disclosure. We combine the valid textual classifications with Huazheng environmental scores to form the dependent-variable panel. We then match this panel to audit firm industry-specialization measures calculated from the full A-share audit market. Finally, we exclude observations with substantial missing financial, governance, or audit data and require nonmissing values for the dependent variable, both main specialization measures, all control variables, firm and year identifiers, and the industry code used in the analysis. These procedures yield a common sample of 8,129 firm-year observations representing 2,669 firms, four years, and 47 industries. All reported specifications use this common sample, so differences across specifications are not attributable to changes in sample composition.

3.2 Variable Measurement

3.2.1 Measure of Greenwashing

We construct the dependent variable by adapting the DeepGreen procedure (Xu et al., 2026). The first stage is a Segmentation Layer. We extract the "Environmental Information" section from each annual report, segment its Chinese text using Jieba, and remove terms contained in the HIT Stopword List. Layer A then identifies environmentally relevant terms in the segmented text. For each term identified as environmentally relevant, we return to the unsegmented report text and retrieve the complete sentence containing that term. Layer B classifies each resulting keyword–sentence pair according to whether it describes substantive or symbolic environmental disclosure. We implement the two LLM classification layers using Qwen 3.6-32B, a 32-billion-parameter model from the Qwen3 family (Yang et al., 2025). Substantive disclosure identifies a concrete environmental action undertaken by the firm or its subsidiaries, such as implementing a project, adopting a technology, or taking a specific pollution-control measure. Symbolic disclosure

includes general aspirations, slogans, policy restatements, and other statements that do not identify a concrete corporate action.

For each firm-year, we aggregate the Layer B classifications. Let $S_{i,t}$ denote the number of keyword–sentence pairs classified as substantive and $P_{i,t}$ the number classified as symbolic. The green implementation ratio is

$$GI_{i,t} = \frac{S_{i,t}}{S_{i,t} + P_{i,t}}.$$

The counts refer to keyword–sentence pairs rather than unique sentences; consequently, a sentence containing multiple environmental keywords may contribute more than one classified pair. Firm-years without valid classified pairs, or with unresolved technical failures, are not coded as zero disclosure.

Our primary ~~continuous~~ measure compares environmental communication with external environmental performance. Firms in different industries may have systematically different opportunities, technologies, regulatory exposure, and disclosure practices; standardizing across the pooled sample could therefore confound relative greenwashing with cross-industry differences. Following the recommendation to benchmark the components of firm-level greenwashing within industries (Lublóy et al., 2025), we standardize GI and the Huazheng environmental score within each industry-year group. The industry classification is the same as that used for audit firm industry specialization. For each component, the z-score is calculated using the corresponding industry-year mean and population standard deviation ($ddof = 0$). We define

$$Greenwashing_{i,t} = GI_z_{i,t} - ESGE_z_{i,t}.$$

A higher value indicates that a firm's substantive share of green communication is high relative to its industry-year peers while its externally assessed environmental performance is comparatively low, and therefore represents a larger communication–performance gap. As an alternative measure, we construct a binary indicator equal to one when GI is strictly above its industry-year mean and the Huazheng environmental score is strictly below its industry-year mean, and zero otherwise.

Adapting the human-validation approach in DeepGreen (Xu et al., 2026), we assess the reliability of the semantic classifications through independent human review. We recruited three students with backgrounds in ESG and finance, each of whom independently evaluated the same validation observations. The human reference label was determined by majority vote, with an observation assigned a value of one when at least two annotators classified it as one and zero otherwise. The Layer A validation sample contains 1,000 observations, stratified into equal numbers of model-predicted positive and negative observations within each of ten review rounds. The Layer B validation sample contains 500 observations, balanced between model-predicted substantive and symbolic classifications.

Figures 1 and 2 report Accuracy, Precision, Recall, F1, and MCC relative to both the human majority label and the subset on which all three annotators agree, together with the annotators' pairwise Cohen's kappa statistics. Relative to the human majority label, Layer A achieves accuracy of 0.978 and an F1 score of 0.978, while Layer B achieves accuracy of 0.954 and an F1 score of 0.953. The annotators fully agree on 96.8 percent of Layer A observations and 93.4 percent of Layer B observations. These results indicate high agreement both between the model and human reviewers and among the reviewers themselves.

3.2.2 Measures of Audit Firm Industry Specialization

Our explanatory variable measures specialization at the audit-firm level. Audit firms are identified by the stable CSMAR institution identifier. Following the market-share approach used in prior industry-expertise research (Cai and Xian, 2007; Krishnan, 2003), we calculate specialization from the full A-share audit market.

Let $Revenue_{i,t}$ be client operating revenue, a index audit firms, and j index industries. Our primary continuous measure, IMS, is

$$IMS_{a,j,t} = \frac{\sum_{i \in (a,j,t)} Revenue_{i,t}}{\sum_{i \in (j,t)} Revenue_{i,t}}.$$

The numerator is the operating revenue of all valid A-share clients audited by firm a in industry j and year t, and the denominator is the operating revenue of all valid A-share firms in that industry-year. We include only observations with positive operating revenue and nonmissing audit-firm and industry identifiers. The resulting audit-firm–industry–year measure is matched to each client using year, audit-firm identifier, and this mixed industry code. An unmatched specialization observation remains missing.

As an alternative continuous measure, IPS captures the importance of industry j within audit firm a's annual client portfolio:

$$IPS_{a,j,t} = \frac{\sum_{i \in (a,j,t)} Revenue_{i,t}}{\sum_{i \in (a,t)} Revenue_{i,t}}.$$

The numerator is the same audit-firm–industry–year client revenue used in IMS, whereas the denominator is the operating revenue of all valid A-share clients audited by firm a in year t. Thus, IMS measures the audit firm's share of an industry's audit market, while IPS measures the industry's importance in the audit firm's client portfolio. Following the 10 percent specialization threshold used in Chinese audit-market research (Cai and Xian, 2007), we define IMS_D as one when IMS is at least 10 percent and zero otherwise, and define IPS_D analogously as one when IPS is at least 10 percent and zero otherwise.

3.2.3 ~~Measures of~~ Control Variables

We include financial, governance, and audit controls that may be associated with firms' environmental disclosure and performance or with the selection of specialist auditors (Marquis et al., 2016; Reichelt and Wang, 2010; DeFond and Zhang, 2014). Financial controls comprise firm size ($Size$), measured as the natural logarithm of total assets; leverage (Lev), measured as total liabilities divided by total assets; return on assets (ROA), measured as net profit divided by total assets; operating cash flow (CFO), measured as operating net cash flow divided by total assets; sales growth ($Growth$), measured as current-year operating revenue divided by prior-year operating revenue minus one; and firm age ($FirmAgeLog$), measured as the natural logarithm of one plus firm age. Growth is first calculated using consecutive-year revenue observations in the full-market financial data and is then matched to the research panel, which avoids creating artificial lags within the selected sample.

Governance controls comprise state ownership (SOE), an indicator for state-controlled firms; CEO–chair duality ($Dual$); board size ($BoardSize$), measured as the natural logarithm of the number of

directors; and the proportion of independent directors (*IndependentDirectorRatio*). Audit controls comprise an indicator for a Big Four auditor ($Big4$); audit fees ($LnAuditFee$), measured as the natural logarithm of total audit fees; auditor tenure ($AuditorTenure$); and an indicator for an audit-firm change ($IsChangeFirm$). These audit controls help distinguish industry specialization from general audit-firm brand, audit effort, accumulated client knowledge, and changes in the auditor–client relationship (DeFond and Zhang, 2014). Values that do not satisfy the required positive-denominator or logarithm conditions, observations with nonconsecutive years for growth, and undefined or infinite ratios are treated as missing.

3.3 Empirical Model

We estimate the following pooled OLS model:

$$Greenwashing_{i,t} = \alpha + \beta Specialization_{i,t} + \gamma' Controls_{i,t} + YearFE_t + IndustryFE_j + \varepsilon_{i,t}.$$

The dependent variable is the continuous greenwashing measure. *Specialization* is alternatively IMS and IMS_D, and *Controls* contains the financial, governance, and audit variables defined above. The primary specification includes year and industry fixed effects, using the industry classification described above, and standard errors clustered by firm. H1 predicts $\beta < 0$. We estimate successive specifications containing only specialization, then adding financial controls, governance controls, audit controls, and finally year and industry fixed effects. We also estimate the complete model with industry-clustered standard errors. All specifications use the same 8,129 firm-year observations.

4 Empirical Results

4.1 Descriptive Statistics

Table 1 presents descriptive statistics for the common regression sample. Greenwashing has a mean of 0.002 and a standard deviation of 1.331. The mean IMS is 0.066, while the mean IMS_D indicates that 22.9 percent of firm-years are audited by firms meeting the 10 percent threshold. State-controlled firms comprise 22.7 percent of the sample, Big Four clients 4.3 percent, and audit-firm changes 11.7 percent.

4.2 Correlation Analysis

Table 1 reports the ~~complete~~ correlation ~~matrix~~. Pearson correlations are used when both variables are continuous; Spearman correlations are used when either variable is binary. Greenwashing is negatively correlated with IMS (Pearson = -0.060) and IMS_D (Spearman = -0.044). The 0.728 correlation between IMS and IMS_D is expected because the latter is constructed from the former, so they enter separate models. The largest absolute correlation among controls is 0.668 between Size and LnAuditFee.

4.3 Baseline Regression Results

Tables 2 and 3 report the baseline results. The coefficient on IMS is negative throughout. In the primary specification, column (5), it is -0.8739 (standard error 0.3150), significant at 1 percent. The IMS_D coefficient is likewise consistently negative and equals -0.1129 (0.0523) in column (5), significant at 5 percent. Conditional on the controls and fixed effects, clients of firms meeting

the specialization threshold therefore have a greenwashing measure lower by approximately 0.113 on average. With industry-clustered standard errors in column (6), IMS and IMS_D remain significant at 5 percent. According to the findings, these results provide support for H1.

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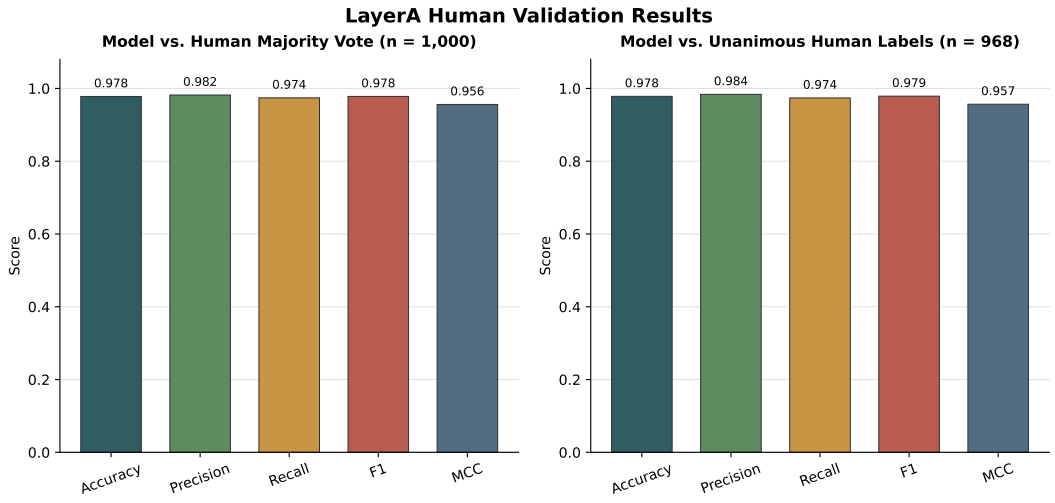
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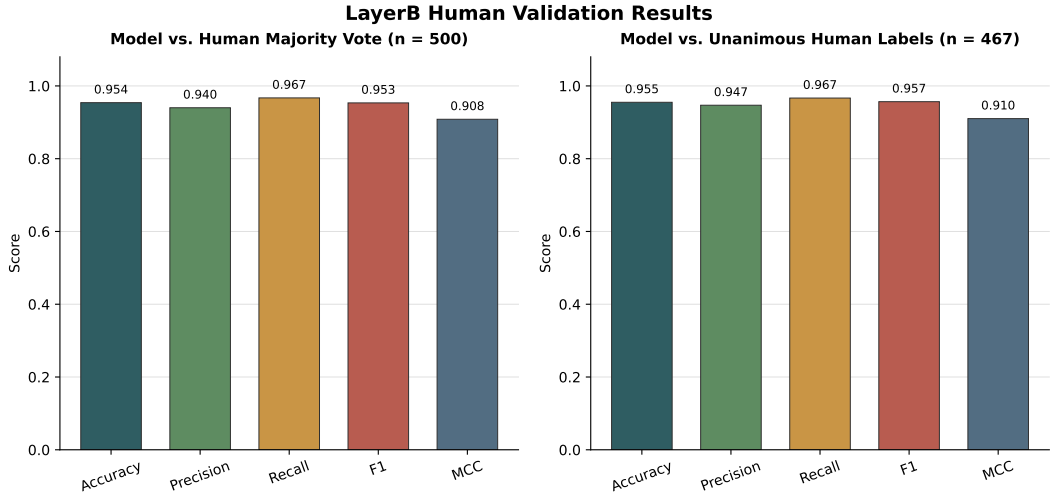
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Human Annotator Pair	Cohen's kappa
Annotator 1 vs. Annotator 2	0.966
Annotator 1 vs. Annotator 3	0.970
Annotator 2 vs. Annotator 3	0.936

Figure 1: Layer A human-validation results. The bar charts compare model classifications with majority-vote and unanimous human labels; the table reports pairwise Cohen's kappa.



Human Annotator Pair	Cohen's kappa
Annotator 1 vs. Annotator 2	0.900
Annotator 1 vs. Annotator 3	0.968
Annotator 2 vs. Annotator 3	0.868

Figure 2: Layer B human-validation results. The bar charts compare model classifications with majority-vote and unanimous human labels; the table reports pairwise Cohen's kappa.

Table 1: Descriptive Statistics and Correlation Matrix

Panel A: Descriptive Statistics

Variable	N	Mean	Std. Dev.	25th Pctl.	Median	75th Pctl.	Max
Greenwashing	8,129	0.002	1.331	-0.809	-0.055	0.902	5.026
IMS	8,129	0.066	0.078	0.014	0.040	0.094	0.704
IMS_D	8,129	0.229	0.420	0	0	0	1
Size	8,129	22.168	1.166	21.342	21.955	22.802	27.535
Lev	8,129	0.405	0.203	0.244	0.391	0.545	1.939
ROA	8,129	0.022	0.076	0.004	0.028	0.057	0.759
CFO	8,129	0.045	0.071	0.009	0.043	0.081	0.839
Growth	8,129	0.110	0.742	-0.082	0.052	0.201	33.767
FirmAgeLog	8,129	2.166	0.812	1.609	2.398	2.708	3.497
SOE	8,129	0.227	0.419	0	0	0	1
Dual	8,129	0.364	0.481	0	0	1	1
BoardSize	8,129	2.066	0.197	1.946	2.197	2.197	2.773
IndependentDirectorRatio	8,129	0.382	0.054	0.333	0.375	0.429	0.714
Big4	8,129	0.043	0.203	0	0	0	1
LnAuditFee	8,129	13.950	0.562	13.580	13.864	14.279	17.127
AuditorTenure	8,129	7.728	5.877	3	6	11	34
IsChangeFirm	8,129	0.117	0.322	0	0	0	1

Panel B: Pairwise Correlations

Variables	Greenwashing	IMS	IMS_D	Size	Lev	ROA	CFO	Growth	FirmAge	SOE	Dual	BoardSize	IndDirector	Big4	AuditFee	Tenure	AuditorChange
Greenwashing	1.000																
IMS	**-0.060**	1.000															
IMS_D	**-0.044**	**0.728**	1.000														
Size	**-0.083**	**0.131**	**0.071**	1.000													
Lev	0.007	**-0.027**	**-0.025**	**0.410**	1.000												
ROA	**-0.030**	**0.101**	**0.082**	**0.105**	**-0.350**	1.000											
CFO	**-0.036**	**0.114**	**0.098**	**0.130**	**-0.158**	**0.423**	1.000										
Growth	-0.016	0.013	0.018	**0.032**	**0.043**	**0.153**	**0.049**	1.000									
FirmAge	0.014	**-0.023**	**-0.045**	**0.430**	**0.322**	**-0.145**	0.009	**-0.027**	1.000								
SOE	-0.015	**-0.039**	-0.018	**0.282**	**0.213**	**-0.080**	**-0.034**	**-0.029**	**0.381**	1.000							
Dual	-0.014	0.022	0.007	**-0.161**	**-0.095**	0.009	-0.008	**0.022**	**-0.209**	**-0.297**	1.000						
BoardSize	-0.008	**0.037**	0.021	**0.238**	**0.123**	**0.034**	**0.032**	0.018	**0.164**	**0.303**	**-0.200**	1.000					
IndDirector	-0.020	0.006	-0.011	**-0.057**	**-0.044**	-0.014	0.006	-0.012	**-0.041**	**-0.145**	**0.155**	**-0.604**	1.000				
Big4	**-0.067**	**0.099**	**0.067**	**0.191**	**0.057**	**0.049**	**0.095**	0.009	**0.045**	**0.057**	**-0.045**	**0.067**	0.011	1.000			
AuditFee	**-0.078**	**0.143**	**0.075**	**0.668**	**0.383**	**-0.082**	**0.082**	0.018	**0.339**	**0.113**	**-0.071**	**0.139**	-0.015	**0.262**	1.000		
Tenure	**0.052**	**0.103**	**0.084**	**0.089**	0.009	**0.062**	**0.072**	-0.004	**0.178**	**-0.206**	**0.043**	**0.040**	**-0.038**	**-0.072**	**0.056**	1.000	
AuditorChange	0.003	**-0.075**	**-0.029**	**0.044**	**0.051**	**-0.071**	**-0.031**	**-0.028**	**0.105**	**0.166**	**-0.050**	**0.046**	**-0.024**	0.019	**0.037**	**-0.559**	1.000

Notes: Panel A reports descriptive statistics. Panel B reports Pearson correlations when both variables are continuous and Spearman correlations when either variable is binary. Coefficients in bold are significant at the 5 percent level (two-tailed). All statistics use 8,129 observations.

Table 2: Audit Firm Industry Specialization and Corporate Greenwashing: Panel A: IMS

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: IMS						
IMS	-1.0238***	-0.7200**	-0.7305**	-0.7477**	-0.8739***	-0.8739**
	(0.3157)	(0.3170)	(0.3159)	(0.3175)	(0.3150)	(0.4049)
Size		-0.1323***	-0.1292***	-0.0954***	-0.1079***	-0.1079**
		(0.0261)	(0.0265)	(0.0304)	(0.0328)	(0.0449)
Lev		0.2683**	0.2707**	0.2898**	0.2720*	0.2720*
		(0.1334)	(0.1333)	(0.1337)	(0.1404)	(0.1505)
ROA		0.3020	0.2937	0.1341	0.1043	0.1043
		(0.2824)	(0.2820)	(0.2813)	(0.2864)	(0.3031)
CFO		-0.3105	-0.3144	-0.2587	-0.2892	-0.2892
		(0.2565)	(0.2563)	(0.2546)	(0.2580)	(0.2419)
Growth		-0.0243**	-0.0246**	-0.0228*	-0.0225*	-0.0225
		(0.0124)	(0.0125)	(0.0127)	(0.0131)	(0.0155)
FirmAgeLog		0.0856***	0.0904***	0.0594*	0.0541	0.0541*
		(0.0312)	(0.0324)	(0.0336)	(0.0349)	(0.0328)
SOE			-0.0622	-0.0394	-0.0549	-0.0549
			(0.0645)	(0.0655)	(0.0671)	(0.0669)
Dual			-0.0350	-0.0374	-0.0308	-0.0308
			(0.0485)	(0.0484)	(0.0489)	(0.0474)
BoardSize			-0.0387	-0.0389	-0.0443	-0.0443
			(0.1550)	(0.1530)	(0.1546)	(0.1518)
IndependentDirectorRatio			-0.6440	-0.5226	-0.4995	-0.4995
			(0.5511)	(0.5478)	(0.5576)	(0.5016)
Big4				-0.1987	-0.2094	-0.2094*
				(0.1295)	(0.1302)	(0.1116)
LnAuditFee				-0.0850	-0.0796	-0.0796
				(0.0610)	(0.0634)	(0.0656)
AuditorTenure				0.0152***	0.0142***	0.0142***
				(0.0043)	(0.0044)	(0.0040)
IsChangeFirm				0.1214**	0.1164**	0.1164***
				(0.0517)	(0.0528)	(0.0346)
Intercept	0.0697**	2.6991***	2.9732***	3.2970***	3.4896***	3.4896***
	(0.0306)	(0.5298)	(0.6778)	(0.8394)	(0.8738)	(1.2847)
Observations	8,129	8,129	8,129	8,129	8,129	8,129
Adjusted R^2	0.0035	0.0127	0.0131	0.0183	0.0187	0.0187
Year fixed effects	No	No	No	No	Yes	Yes
Industry fixed effects	No	No	No	No	Yes	Yes

Notes: The dependent variable is continuous Greenwashing. Columns (1)–(5) use firm-clustered standard errors; column (6) uses industry-clustered standard errors. Coefficients are reported above standard errors in parentheses. ***, **, and * indicate two-tailed significance at 1, 5, and 10 percent.

Table 3: Audit Firm Industry Specialization and Corporate Greenwashing: Panel B: IMS_D

	(1)	(2)	(3)	(4)	(5)	(6)
Panel B: IMS_D						
IMS_D	-0.1471***	-0.0992**	-0.1003**	-0.1030**	-0.1129**	-0.1129**
	(0.0511)	(0.0506)	(0.0506)	(0.0506)	(0.0523)	(0.0556)
Size		-0.1358***	-0.1328***	-0.0969***	-0.1124***	-0.1124**
		(0.0259)	(0.0264)	(0.0303)	(0.0327)	(0.0446)
Lev		0.2755**	0.2773**	0.2973**	0.2847**	0.2847*
		(0.1332)	(0.1331)	(0.1334)	(0.1401)	(0.1506)
ROA		0.2929	0.2843	0.1201	0.0869	0.0869
		(0.2824)	(0.2819)	(0.2814)	(0.2870)	(0.3015)
CFO		-0.3351	-0.3375	-0.2769	-0.2998	-0.2998
		(0.2576)	(0.2571)	(0.2555)	(0.2584)	(0.2459)
Growth		-0.0236*	-0.0239*	-0.0220*	-0.0218*	-0.0218
		(0.0123)	(0.0125)	(0.0126)	(0.0129)	(0.0154)
FirmAgeLog		0.0862***	0.0906***	0.0605*	0.0561	0.0561*
		(0.0312)	(0.0324)	(0.0336)	(0.0349)	(0.0327)
SOE			-0.0557	-0.0352	-0.0538	-0.0538
			(0.0644)	(0.0654)	(0.0671)	(0.0669)
Dual			-0.0328	-0.0351	-0.0296	-0.0296
			(0.0486)	(0.0485)	(0.0490)	(0.0476)
BoardSize			-0.0486	-0.0476	-0.0504	-0.0504
			(0.1555)	(0.1534)	(0.1549)	(0.1522)
IndependentDirectorRatio			-0.6783	-0.5561	-0.5193	-0.5193
			(0.5520)	(0.5487)	(0.5575)	(0.5086)
Big4				-0.2059	-0.2213*	-0.2213**
				(0.1296)	(0.1302)	(0.1123)
LnAuditFee				-0.0905	-0.0815	-0.0815
				(0.0608)	(0.0633)	(0.0656)
AuditorTenure				0.0148***	0.0138***	0.0138***
				(0.0044)	(0.0044)	(0.0039)
IsChangeFirm				0.1206**	0.1152**	0.1152***
				(0.0518)	(0.0529)	(0.0351)
Intercept	0.0356	2.7477***	3.0571***	3.4095***	3.5843***	3.5843***
	(0.0260)	(0.5278)	(0.6778)	(0.8382)	(0.8713)	(1.2923)
Observations	8,129	8,129	8,129	8,129	8,129	8,129
Adjusted R^2	0.0020	0.0119	0.0123	0.0176	0.0177	0.0177
Year fixed effects	No	No	No	No	Yes	Yes
Industry fixed effects	No	No	No	No	Yes	Yes

Notes: The dependent variable is continuous Greenwashing. Columns (1)–(5) use firm-clustered standard errors; column (6) uses industry-clustered standard errors. Coefficients are reported above standard errors in parentheses. ***, **, and * indicate two-tailed significance at 1, 5, and 10 percent.